Pearson Edexcel International GCSE

Mathematics A

Modular Unit 1

June 2023 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

29 adaptive question sections   97 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each question section keeps the exact source demand beside practical working space. A source grid, graph, table, or diagram is retained once at a usable size.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 1; the selection contains 97 marks.

Contents

Question 01

4MA1-1H Q5

MODULAR UNIT 1
4 marks

5 The diagram shows a shaded shape *AEBCD* made by removing triangle *AEB* from rectangle *ABCD*

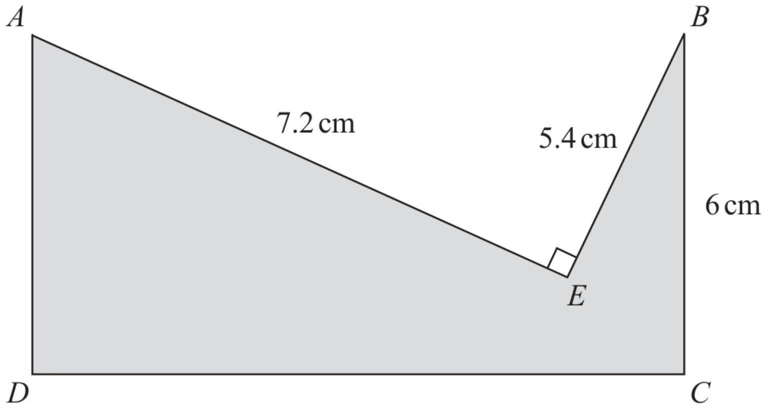


Diagram **NOT** accurately drawn

$AE = 7.2$ cm $\quad BE = 5.4$ cm $\quad BC = 6$ cm $\quad$ angle $AEB = 90°$

Work out the perimeter of the shaded shape.

.... cm

WORKING SPACE

Question 02

4MA1-1H Q6

MODULAR UNIT 1
2 marks

6 (a) Simplify $(2c^4d^7)^3$

....
(2)

WORKING SPACE

Question 03

4MA1-1H Q6

MODULAR UNIT 1

1 marks

(b) Find the value of $5y^0$ where $y > 0$

....

(1)

WORKING SPACE

Question 04

4MA1-1H Q6

MODULAR UNIT 1

2 marks

(c) Factorise fully $16a^2b^3 + 20a^3b$

....

(2)

WORKING SPACE

Question 05

4MA1-1H Q6

MODULAR UNIT 1

3 marks

(d) (i) Factorise $x^2 + 9x - 22$

....

(2)

(ii) Hence solve $x^2 + 9x - 22 = 0$

....

(1)

WORKING SPACE

Question 06

4MA1-1H Q13

MODULAR UNIT 1

3 marks

13 A is the point with coordinates (–5, 12)

B is the point with coordinates (19, –48)

Find an equation of the straight line that passes through the points A and B

....

WORKING SPACE

Question 07

4MA1-1H Q14

MODULAR UNIT 1

3 marks

14 Factorise fully $50g^2 - 18$

....

WORKING SPACE

Question 08

4MA1-1H Q15

MODULAR UNIT 1

3 marks

15 (a) $\sqrt{2} \div \dfrac{8^3}{16^{\frac{3}{2}}} = 2^n$

Work out the value of n

Show your working clearly.

$n =$ ....

(3)

WORKING SPACE

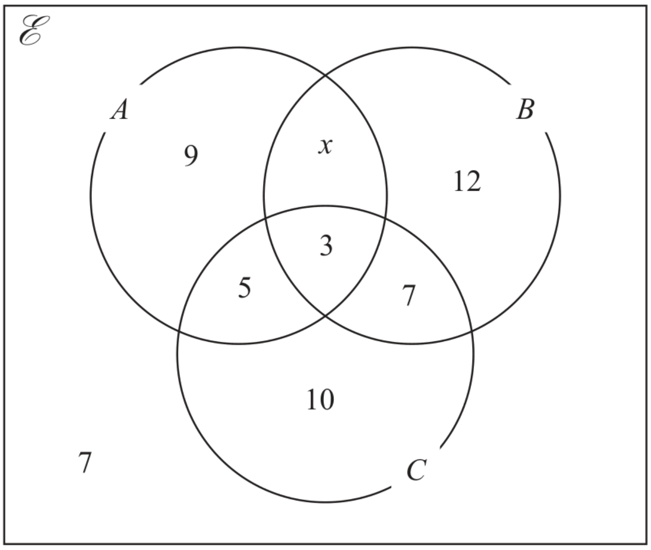
Question 09

4MA1-1H Q16

MODULAR UNIT 1
3 marks

16 The Venn diagram shows a universal set $\mathscr{E}$ and sets A, B and C

The numbers and the letter x represent **numbers** of elements.



Given that $n(A \cup B) = 42$

(a) find the value of x

$x =$ ....

(1)

(b) Find $n(A')$

....

(1)

(c) Find $n(B' \cap C)$

....

(1)

Working space for Question 09

Continue your method

UNIT 1
3 marks

WORKING SPACE

Question 10

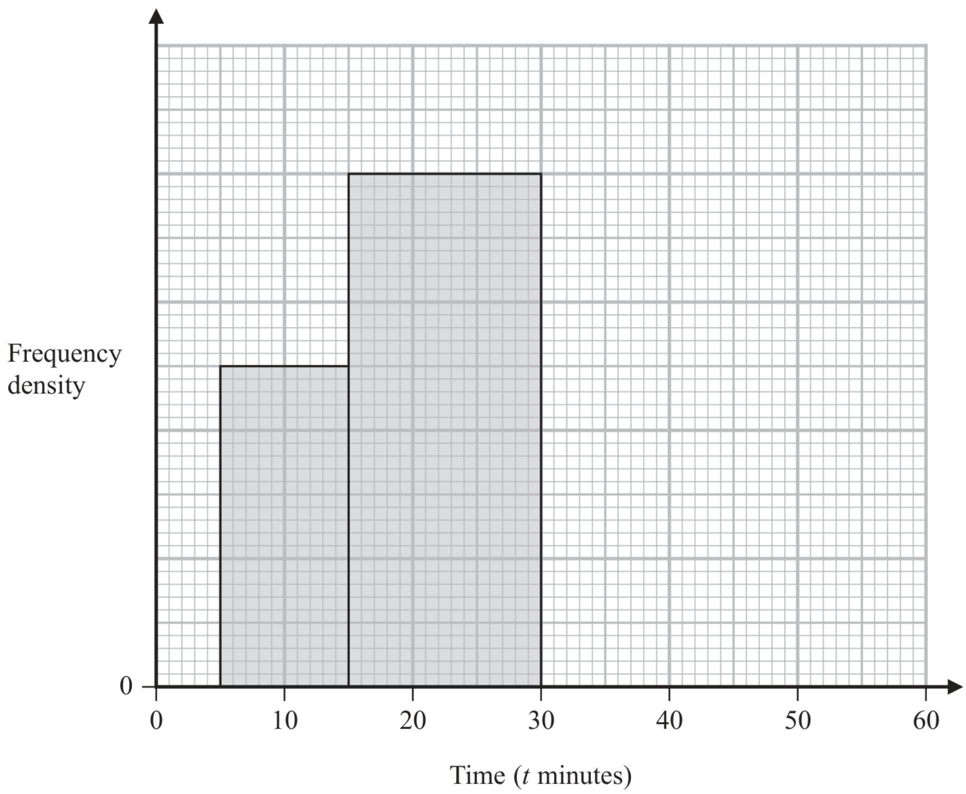
4MA1-1H Q18

MODULAR UNIT 1

4 marks

18 The incomplete table and incomplete histogram give information about the times, in minutes, that 140 people waited at a station for a train.

Time (t minutes)	**Frequency**
$0 < t \leqslant 5$	23
$5 < t \leqslant 15$	
$15 < t \leqslant 30$	
$30 < t \leqslant 40$	18
$40 < t \leqslant 60$	14



Complete the table and the histogram.

Question 11
4MA1-1H Q22

MODULAR UNIT 1
6 marks

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

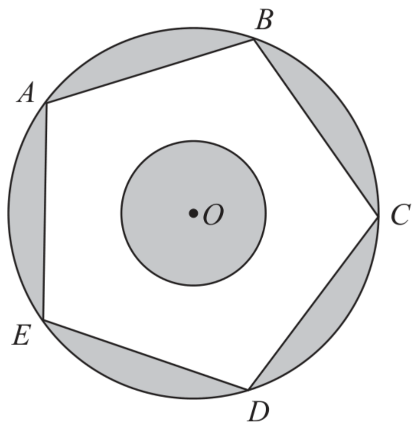


Diagram **NOT** accurately drawn

A, B, C, D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

shaded area = unshaded area

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

.... cm

Working space for Question 11

Continue your method

UNIT 1
6 marks

WORKING SPACE

Question 12

4MA1-2H Q1

MODULAR UNIT 1
3 marks

1 Show that $4\frac{2}{3} \div 1\frac{1}{5} = 3\frac{8}{9}$

WORKING SPACE

Question 13

4MA1-2H Q2

MODULAR UNIT 1
3 marks

2 A biased spinner can land on green or on yellow or on brown or on pink.

The table gives the probabilities that, when the spinner is spun, it will land on green or on yellow or on brown.

Colour	**green**	**yellow**	**brown**	**pink**
Probability	0.32	0.13	0.28	

Timucin spins the spinner 200 times.

Work out an estimate for the number of times the spinner lands on pink.

....

WORKING SPACE

Question 14
4MA1-2H Q3

MODULAR UNIT 1
4 marks

3 *ABCD* is a trapezium.

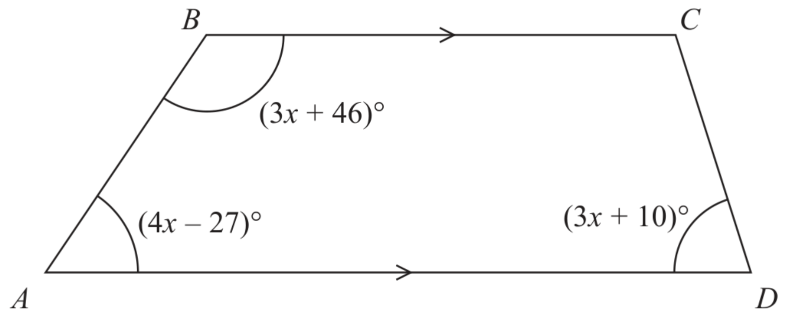


Diagram **NOT** accurately drawn

BC is parallel to *AD*

Find the size of the largest angle inside the trapezium.

.... °

WORKING SPACE

Question 15
4MA1-2H Q4

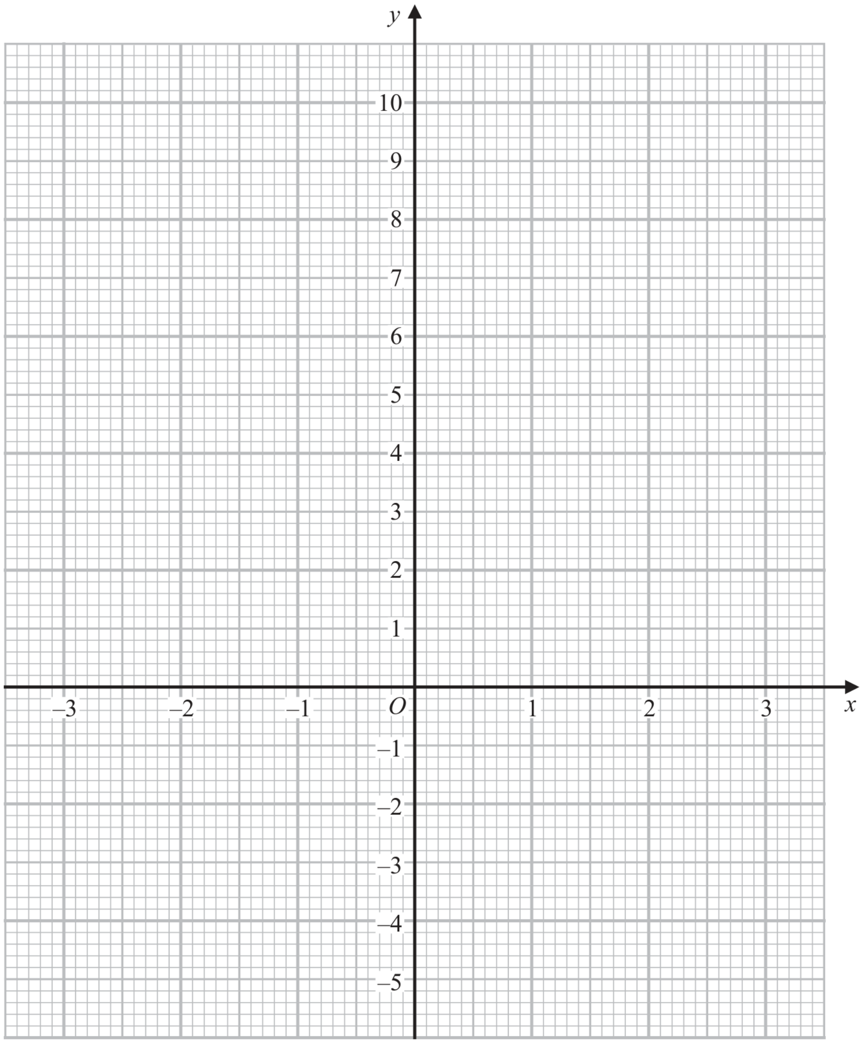
MODULAR UNIT 1
4 marks

4 (a) Complete the table of values for $y = x^2 - x - 4$

x	-3	-2	-1	0	1	2	3
y		2			-4		

(2)

(b) On the grid below, draw the graph of $y = x^2 - x - 4$ for values of x from -3 to 3



(2)

Question 16

4MA1-2H Q9

MODULAR UNIT 1
3 marks

9 Change a speed of 27 kilometres per hour to a speed in metres per second.

.... m/s

WORKING SPACE

Question 17

4MA1-2H Q12

MODULAR UNIT 1
4 marks

12 The diagram shows right-angled triangle *ABD*

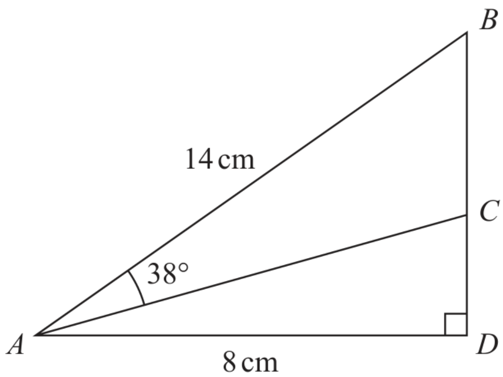


Diagram **NOT** accurately drawn

$AB = 14\text{ cm}$ $AD = 8\text{ cm}$

C is the point on BD such that angle $BAC = 38°$

Work out the length of CD
Give your answer correct to 3 significant figures.

.... cm

WORKING SPACE

Question 18

4MA1-2H Q13

MODULAR UNIT 1

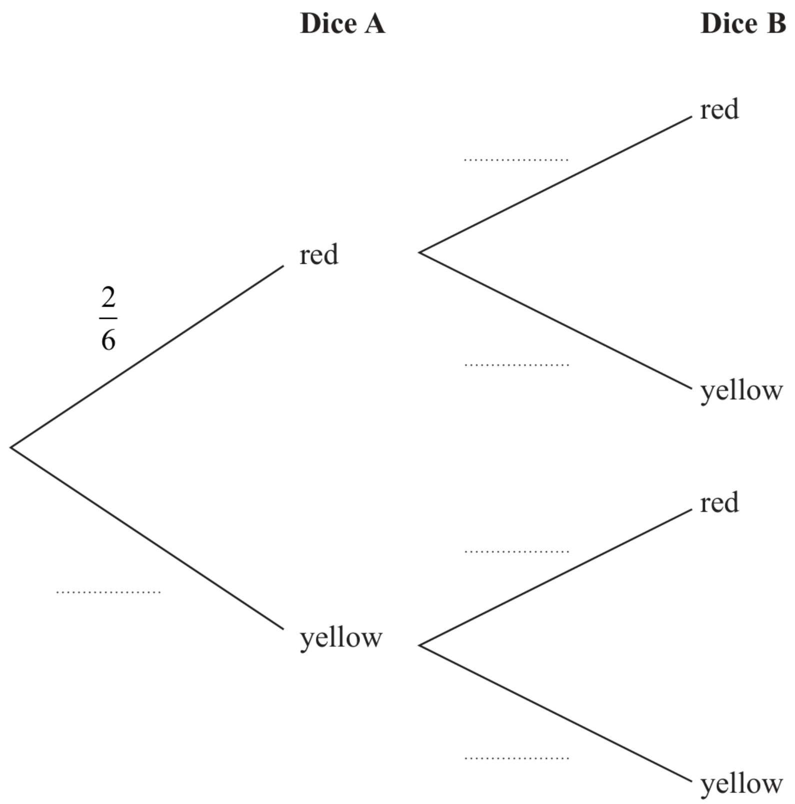
4 marks

13 Narin has two fair 6-sided dice.

Dice **A** has 2 red faces and 4 yellow faces.
Dice **B** has 1 red face and 5 yellow faces.

Narin is going to throw each dice once.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that both dice land on yellow.

....

(2)

Question 19

4MA1-2H Q15

MODULAR UNIT 1
2 marks

15 (a) Simplify fully $\left(32a^{15}\right)^{\frac{3}{5}}$

....

(2)

WORKING SPACE

Question 20

4MA1-2H Q15

MODULAR UNIT 1
2 marks

(b) Express $\left(\frac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

....

(2)

WORKING SPACE

Question 21

4MA1-2H Q15

MODULAR UNIT 1
3 marks

(c) Solve $\dfrac{1-2y}{3} = \dfrac{4}{5} - \dfrac{2y-1}{2}$

Show clear algebraic working.

$y =$ ....

(3)

WORKING SPACE

Question 22
4MA1-2H Q17

MODULAR UNIT 1
3 marks

17 (a) Expand and simplify $(x + 6)(3x - 2)(x + 6)$

....

(3)

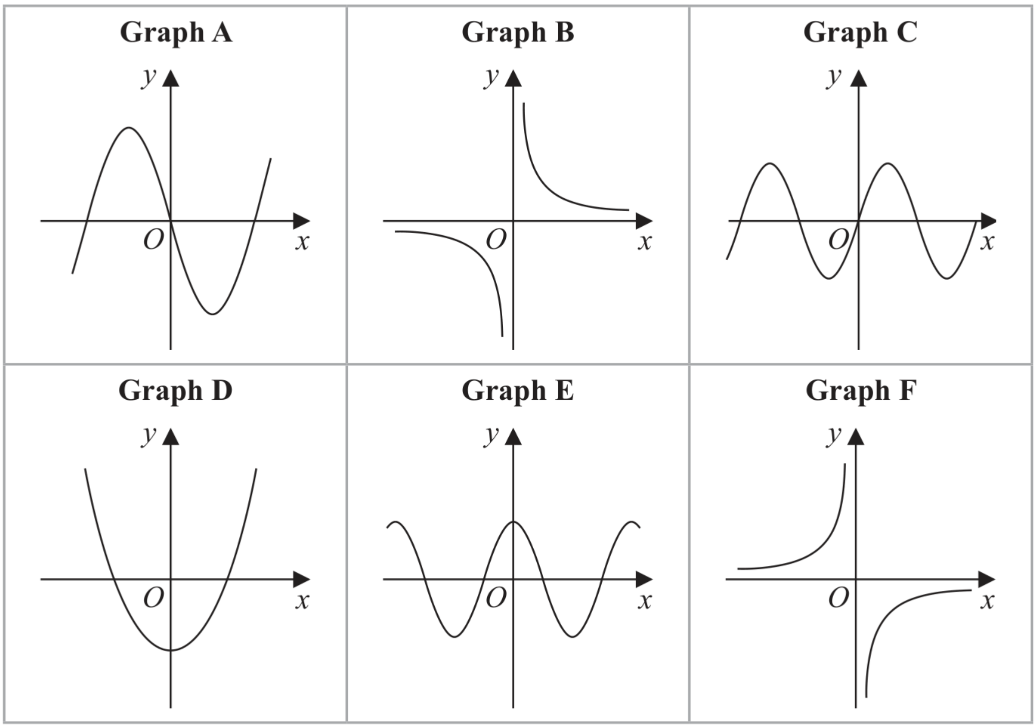
WORKING SPACE

Question 23

4MA1-2H Q18

MODULAR UNIT 1
3 marks

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \sin x$	....
$y = -\dfrac{3}{x}$	....
$y = 4x^3 - 5x$	....

Working space for Question 23

Continue your method

UNIT 1
3 marks

WORKING SPACE

Question 24

4MA1-2H Q19

MODULAR UNIT 1

3 marks

19 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

....

WORKING SPACE

Question 25

4MA1-2H Q20

MODULAR UNIT 1
3 marks

20 There are 12 counters in a bag.

3 of the counters are red
9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

....

WORKING SPACE

Question 26

4MA1-2H Q22

MODULAR UNIT 1
5 marks

22 The diagram shows a triangle *ABC* and a flagpole *BF*

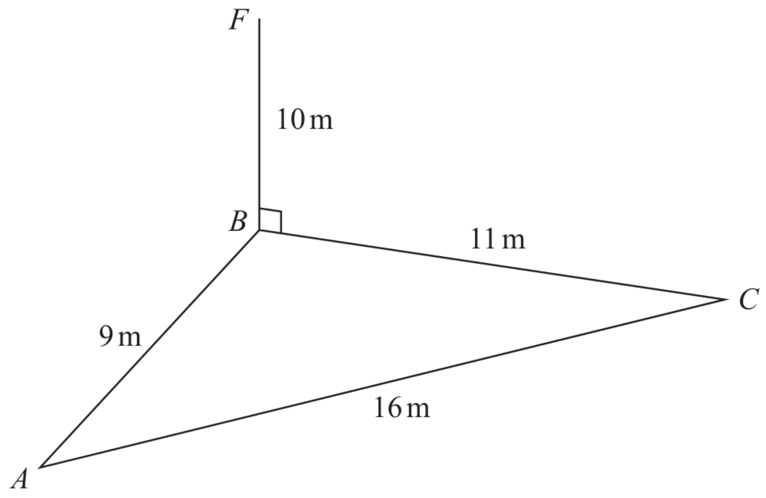


Diagram **NOT** accurately drawn

A, *B* and *C* are points on horizontal ground.

BF is vertical.

$AB = 9\,\text{m}$ $BC = 11\,\text{m}$ $AC = 16\,\text{m}$ $BF = 10\,\text{m}$

D is the point on *AC* such that angle $BDC = 90°$

Work out the size of the angle of elevation of the point *F* from the point *D*
Give your answer correct to one decimal place.

....°

Working space for Question 26

Continue your method

UNIT 1
5 marks

WORKING SPACE

Question 27

4MA1-2H Q23

MODULAR UNIT 1
4 marks

23 The diagram shows a cuboid with a square cross section.

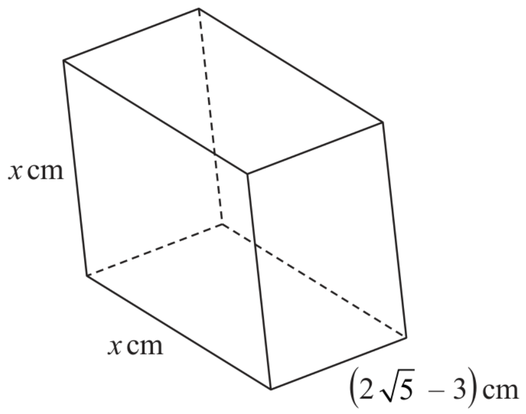


Diagram **NOT** accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\,\text{cm}^3$

Without using a calculator, find the value of x
Give your answer in the form $a + \sqrt{b}$ where a and b are integers.
Show your working clearly.

$x =$ ....

Working space for Question 27

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 28

4MA1-2H Q24

MODULAR UNIT 1
6 marks

24 *ABCD* is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of *D*

(.......................... , ...)

WORKING SPACE

Question 29

4MA1-2H Q25

MODULAR UNIT 1
4 marks

25 A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.
Give your answer in g/cm^3 correct to 2 decimal places.
Show your working clearly.

.... g/cm^3

WORKING SPACE